Quiz Bowl RL Buzzer

Learning when to buzz under incremental clues

Kathleen Weng • Imran Hassan • Ankit Aggarwal
CS234 final project • smoke metrics from evaluation_report.json

One-sentence takeaway

Buzzing becomes an optimal-stopping problem; on this smoke slice, simple baselines remain the strongest anchors.

Problem + design choice

- Clues arrive incrementally, so the system must decide when to buzz — timing matters beyond answer selection.
- Early buzzes risk mistakes; late buzzes waste strategic value.
- A fixed 4-choice setting makes the stopping problem reproducible.

POMDP at a glance

Observation	prefix h_t + choice set C
Actions	WAIT or BUZZ(i)
Goal	buzz when acting beats waiting

Why multiple choice?

- Removes aliasing and grading noise.
- Keeps the gold answer inside the action space.
- Allows cleaner baseline and control comparisons.
- Uses explicit anti-artifact distractor checks.

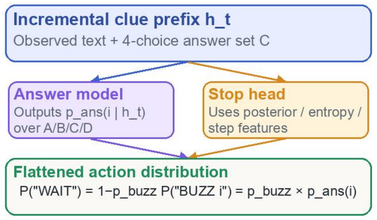
alias collisions | token overlap | length ratio | question overlap

Metrics glossary

S_q	QANTA system score; rewards early correct buzzes
acc	buzz accuracy — fraction of buzzes on the gold answer
step	mean buzz step — average prefix index at buzz time
rew	reward-like — composite PPO training signal

Method

Factor answer quality and stop timing into separate components, then combine into WAIT / BUZZ(i).



1 Build MC prefixes | 2 Run baselines | 3 Warm-start stop | 4 PPO fine-tune

Conclusions

- With a fixed answer set, buzzing becomes an optimal-stopping problem over growing evidence.
- The Markov example shows posterior mass concentrating before the final giveaway.

References

- Rodriguez et al. (2019). Quizbowl and incremental QA.
- QANTA (2024). System score S_q for buzzing.
- Sung et al. (2025). ADVSCORE.

Repo: github.com/hass0114/qanta-buzzer

Key example: the Markov tossup

Example tossup: QBReader 2022 ARCADIA 08/11

Full tossup — pyramidal clue progression (8 runs)

1. Given a set of nodes **named for this person**, a node is conditionally independent from the rest of a Bayesian network.
2. The Baum-Welch algorithm is used to train a type of model **named for this person** that is used for multiple sequence alignment.
3. The initial "burn-in" states of a process **named for this person** are discarded in methods like Gibbs sampling.
4. The Metropolis-Hastings algorithm approximates an unknown distribution as the (*) stationary distribution of a process **named for this person**.
5. Monte Carlo methods often use processes **named for this person** that have stochastic transition matrices.
6. Dynamic programming is used to decode "hidden" models **named for this person**.
7. The next state of a random process **named for this person** depends only on the current state.
8. For 10 points, what Russian mathematician names a type of **memoryless "chain?"**

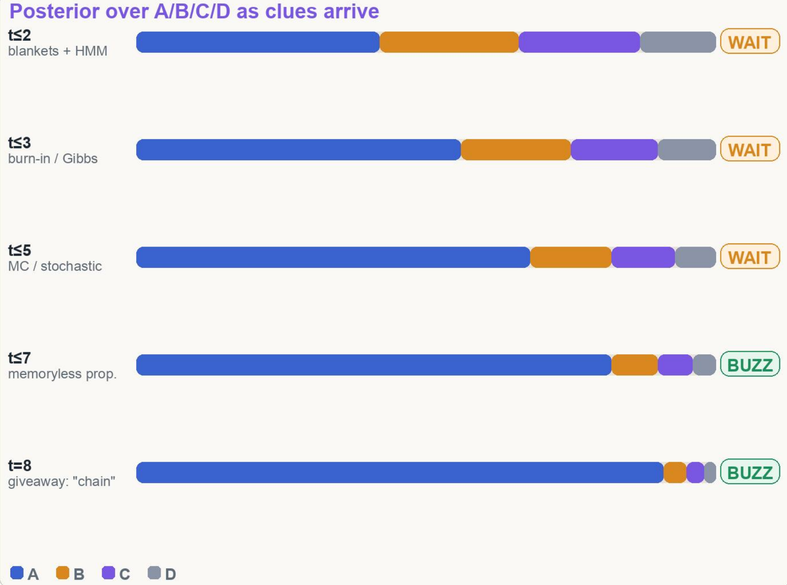
ANSWER: Andrey Andreyevich Markov

Each sentence repeats the oblique referent "named for this person"; the final giveaway says "memoryless chain." Under MC framing, the core challenge shifts to timing the buzz.

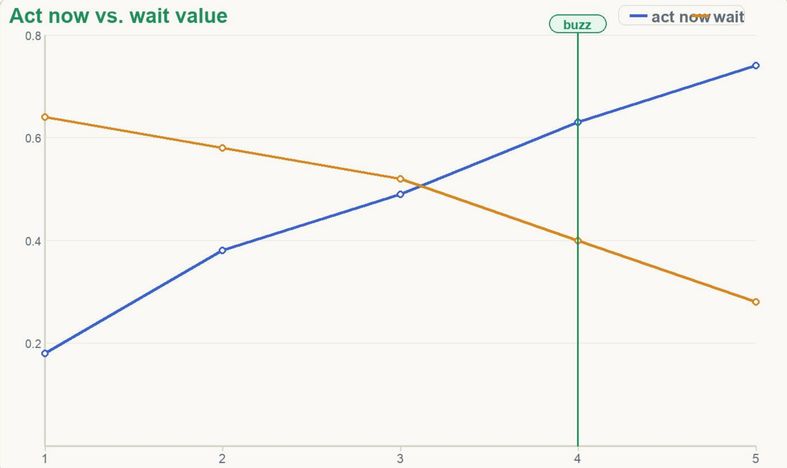
Fixed answer set

A Andrey Andreyevich Markov | B Leonhard Euler | C Carl Friedrich Gauss | D Augustin-Louis Cauchy

Posterior over A/B/C/D as clues arrive

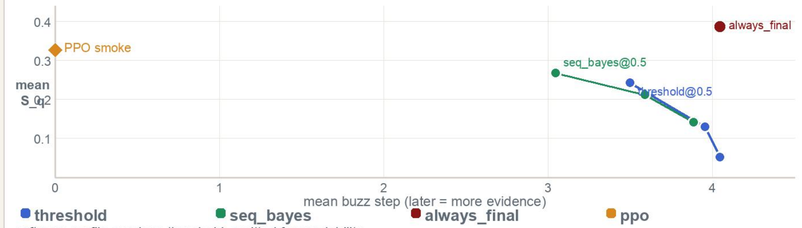


Act now vs. wait value



Smoke snapshot (sanity-check)

n = 44. Smoke-test sanity checks only.



softmax_profile overlaps threshold, omitted for readability.

late baseline	best non-final	ppo smoke
S_q = 0.386	S_q = 0.267	S_q = 0.326
acc = 38.6%	acc = 38.6%	acc = 34.1%
step = 4.05	step = 3.05	step = 0.00
		rew = -0.476

Results + controls

Smoke takeaway: always_final best S_q (0.386); sequential_bayes@0.5 strongest non-final (0.267); PPO reaches 0.326 but buzzes at step 0.0.

Choices only
9.1%
chance = 25.0% • n = 11
Answer options alone do not solve the task.

Shuffle clues
ΔS_q = -0.007
full = 0.243 • shuffled = 0.237
Very small movement on the smoke slice.

Alias substitution
ΔS_q = +0.000
full = 0.243 • alias = 0.243
Essentially unchanged in this smoke report.

Interpret with caution
Choices-only underperforms, but shuffle and alias barely move S_q. These controls are suggestive but inconclusive.

The framing exposes gaps in timing, calibration, and artifact checking that remain open.

Limitations + next checks

- The smoke slice is small: full_eval uses n = 44 and choices-only uses n = 11.
- PPO still shows degenerate timing, so reward design needs another pass.

Poster: content from the bundled slide deck and smoke report; all numerical claims from evaluation_report.json; smoke results are sanity-check snapshots only.